

Continuous Assessment Card

Department of Physiology----- Medical college-----
 Students name----- Roll no.-----
 Session ----- Year ----- Batch -----
 Date of starting ----- Date of ending -----

Card 1: (Cellular Physiology & Blood)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Definition, goal & importance of physiology. Homeostasis: definition, major functional systems, control systems of the body	10		
2.	The cell: functions of cell membrane & cell organelles.	10		
3.	The cell membrane transport: active & passive transport, exocytosis & endocytosis. Intercellular communications	10		
4.	Membrane potential: definition and basic physics of membrane potential. Resting membrane potential Nerve Action potential & propagation of action potential.	10		
5.	Neuromuscular junction & transmission of impulse from nerve ending to the muscle fibre. Muscle contraction & relaxation.	10		
6.	Composition & functions of blood, Plasma proteins: Origin, normal values, properties & functions.	10		
7.	RBC: normal count, morphology, functions, erythropoiesis, fate of RBC. Hemoglobin: synthesis, types, functions. Red blood cell indices. Anaemia: definition & classification Polycythemia: definition & type. Jaundice: definition & classification	10		
8.	WBC: classification with normal count, morphology, development, properties & functions. leucocytosis, leucopenia .	10		
9.	Platelets: normal count, morphology, functions & development. Hemostasis: definition & events Coagulation: definition, blood clotting factors. Mechanism of coagulation & fibrinolysis. Anticoagulant: name, mode of action. Bleeding disorder: thrombocytopenic purpura & hemophilia. Tests for bleeding disorder: bleeding time, coagulation time and prothrombin time.	10		
10.	Blood grouping: ABO & Rh system, hazards of blood transfusion & Rh incompatibility.	10		

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Card 2: (Cardiovascular Physiology)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Properties of cardiac muscle. Junctional tissues of the heart. Generation of cardiac impulse & its conduction in the heart.	10		
2.	Cardiac cycle: definition, events, pressure & volume changes during different phases of cardiac cycle. Heart sounds : type, characteristics and their significances ECG : definition, principles and interpretations Heart block: definition and types.	10		
3.	Functional classification of blood vessels, interrelationship among pressure, flow & resistance. Local & humoral control of blood flow in the tissues. Exchange of fluid through the capillary membrane.	10		
4.	SV, EDV, EF, ESV: definition & factors affecting them. Cardiac output : definition, measurement, regulation and factors affecting cardiac output. Venous return: definition & factors affecting. Pulse: definition, characteristics	10		
5.	Peripheral resistance: definition & factors affecting. Blood pressure: definition, types, measurement & regulation of arterial blood pressure.	10		
6.	Circulatory adjustment during muscular exercise Heart rate: factors affecting & regulation Cardiac arrhythmias : tachycardia, bradycardia. Shock: definition, classification. Physiological basis of compensatory mechanism of circulatory shock.	10		

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Card 3: (Respiratory Physiology)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Respiration: definition, mechanism. Pulmonary & Alveolar ventilation. Pulmonary volumes and capacities(spirometry) Dead space: physiological & anatomical Lung function tests : name & significance	10		
2.	Composition of atmospheric, alveolar, inspired and expired air. Respiratory unit and respiratory membrane. Diffusion of Gases through the respiratory membrane. Peculiarities of pulmonary circulation Ventilation -perfusion ratio.	10		
3.	Transport of Oxygen & Carbon dioxide in blood. Oxy-hemoglobin dissociation curve. Bohr effect, Haldane effect & Chloride shift.	10		
4.	Respiratory centers: name, location & functions. Nervous & chemical regulation of respiration. Regulation of respiration during exercise.	10		
5.	Hypoxia: definition, types Cyanosis: definition & types. Definition of dyspnea, hypercapnea & periodic breathing.	10		

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Card 4 : (Gastrointestinal Physiology & Renal physiology)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (Signature & Date)
1.	Physiological anatomy of gastrointestinal (GI) tract. Enteric nervous system. Local hormones of GIT: name, functions & regulation of secretion	10		
2.	Movements of the GIT. GI reflexes. Functions of stomach, small intestine and large intestine	10		
3.	Kidney: functions of kidneys. Nephron: Types, parts and structure. Renal circulation: peculiarities with functional importance.	10		
4.	Urine formation Glomerular filtration, measurement and determinants of GFR, Autoregulation of renal blood flow and GFR.	10		
5.	Reabsorption and secretion by the renal tubules Definition of T_m , Renal threshold, tubular load, plasma load, plasma clearance and diuresis	10		
6.	Mechanism of formation of concentrated & dilute urine.	10		
7.	Micturition reflex Abnormalities of micturition	10		

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Card 5 : (Endocrine Physiology)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks
1.	Endocrine glands: name Hormones: definition, classification, mechanism of action, regulation of secretion	10		
2.	Hypothalamic hormones. Pituitary hormones (anterior & posterior): name, functions and their control by the hypothalamus and disorders (Dwarfism, gigantism, acromegaly & hypopituitarism and diabetes insipidus)	10		
3.	Thyroid hormones: biosynthesis, transport, functions, regulation of secretion, disorders (Hypothyroidism hyperthyroidism, Cretinism, Myxoedema and goitre). .	10		
4.	Parathyroid hormone: functions, mechanism of action & regulation of secretion. Calcium homeostasis.	10		
5.	Adrenocortical hormones: name, functions , mechanism of action , regulation of secretion & disorders (Addison's disease, Cushing's Syndrome, Conn's disease).	10		
6.	Hormones of Islets of Langerhan's of pancreas: functions , mechanism of action, regulation of secretion. Pathophysiology of insulin deficiency.	10		

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Card 6: (Physiology of Reproduction)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks
1.	Introduction to reproductive physiology, sex determination & sex differentiation. Puberty Functional anatomy of male reproductive system. Secondary sex characteristics of male Gonad : structure and functions of testes. Testosterone: functions, Spermatogenesis: steps & hormonal control.	10		
2.	Functional anatomy of female reproductive system. Secondary sex characteristics of female Gonad : structure and functions of ovaries. Ovarian hormones Oestrogen and progesterone: functions Ovulation: definition, mechanism & hormonal control. Indicators of ovulation Menstrual cycle: definition & hormonal control. Ovarian and endometrial cycle with their hormonal regulation. Definition of menstruation, menarche & menopause.	10		
3.	Placental hormones: name & functions. Mammogenesis: hormonal influence for mammogenesis & lactation	10		

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Card 7: (Neurophysiology & special senses)

Sl. No	Name of item	Full Marks	Marks Obtained	Remarks & signature
1.	Functional organization and functions of major levels of central nervous system(CNS). Neuron: definition, parts, types Nerve fiber: classification, properties, effects of injury to the nerve fiber Synapse: physiological anatomy, type, properties & synaptic transmission Neurotransmitters: definition, types & functions	10		
2.	Sensory systems of the body: Sensory receptor: definition, classification, properties, receptor/generator potential. Cerebral cortex : Name and functions of the Brodmann's areas. General/somatic senses: definition and classification. Ascending tracts/sensory pathways – name.(Tract of Gall & Burdach, spinothalamic tract, spinocerebellar tract): origin, course, termination, functions, and effect of lesions.	10		
3.	Reflex: definition, classification, properties. Reflex arc: definition, component stretch reflex, knee jerk, planter response and Withdrawal reflex- with reciprocal innervations & crossed extensor-pathway . Muscle spindle, Golgi tendon organ: definition, physiological anatomy and functions. Muscle tone : definition , function and maintenance.	10		
4.	Descending tracts/ motor pathways- name Pyramidal tract: origin, course, termination, function, effect of lesion. Extrapyramidal tract: name, functions. Upper motor neuron and lower motor neuron : definition, effect of lesion. Spinal cord : effect of hemisection.	10		
5.	Cerebellum: functional division, neuronal circuit, functions, error control mechanism of motor activity & cerebellar disorder, Basal ganglia: functional components, functions & effects of lesions. Thalamus, Reticular formation, Limbic system: functional components and functions.	10		
6.	Hypothalamus: name of the nucleus, functions Body Temperature Normal body temperature, site of measurement, sources of heat gain, channels of heat loss, regulation of body temperature in hot and cold environment.	10		
7.	Autonomic Nervous system: physiological anatomy of sympathetic and parasympathetic system, functions. Alarm or stress response	10		
8.	Vision: physiological anatomy of eye, image formation in the eyes, visual receptors, visual pathway, common refractive errors, accommodation reaction, light reflex, dark and light adaptation. Field of vision, color vision, visual acuity	10		
9.	Hearing: auditory apparatus, receptor, Mechanism of hearing, mechanism of sound transmission and auditory pathway.	10		
10.	Smell: receptor and pathway. Taste: receptors, modalities of taste sensation and pathway.	10		

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Card 8: Physiology Practical

(I hear and I forget, I see and I remember, I do and I understand)

SL NO	Name of experiment	Full Marks	Marks obtained
1	Laboratory equipment. blood sample, collection (venous & capillary) of blood.	10	
2	Preparation & staining of blood film & differential count of WBC with interpretation and analysis of result	10	
3	Determination of total count of WBC with interpretation and analysis of result	10	
4	Determination of total count of RBC with interpretation and analysis of result	10	
5	Estimation of haemoglobin with interpretation and analysis of result	10	
6	Determination of packed cell volume (PCV), Calculation of MCV, MCH & MCHC with interpretation and analysis of result	10	
7	Estimation of ESR by Westergren method with interpretation and analysis of result	10	
8	Determination of bleeding time, clotting time with interpretation and analysis of result	10	
9	Study of morphology and osmotic behavior of RBC with interpretation and analysis of result	10	
10	Determination of ABO & Rh blood groups with interpretation and analysis of result	10	
11	Auscultation of 1 st & 2 nd heart sounds	10	

12	Examination of radial pulse.	10	
13	Measurement of normal blood pressure & effects of exercise on blood pressure.	10	
14	Recording & analysis of 12 leads normal ECG	10	
15	Auscultation of breath sounds	10	
16	Spirometric measurement of lung function test. Determination of FVC, FEV ₁ , FEV ₁ /FVC %, PEFR, MVV with analysis of result.	10	
17	Auscultation of intestinal sound.	10	
18	Elicitation of knee jerk, planter response	10	
19	Recording of oral & axillary temperature & effects of exercise on it	10	
20	Observation of light reflexes and analysis of result	10	
21	Determination of color vision	10	
22	Determination of visual acuity by Snellen's chart.	10	
23	Determination of hearing tests: Rinne and Weber test with interpretation and analysis of result	10	
24	Determination of specific gravity of urine	10	

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